

Feasibility Report

Project CGHL3: Transporting Protein-Based Medicine

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1 Executive Summary

The aim of this feasibility report is to evaluate how achievable the development of a portable insulin-cooling device is. The purpose of the device is to maintain insulin at a safe storage temperature during travel to remote regions with limited infrastructure and resources, particularly within developing countries. Maintaining insulin at a safe storage temperature remains a major challenge in environments where refrigeration is unreliable. The modelling conducted for this report shows that a compact system combining phase-change materials (PCMs), EPS insulation and a low-power peltier module can consistently maintain the required 2–6 °C range even under high ambient temperatures. These findings demonstrate that the core thermal requirements can be met without depending on grid-based electricity.

Initially each team member drafted 3 concept designs, leading to 18 individual ideas. This was then funnelled down to the best 6 using each member's best design. Each of the proposed concepts retained some core elements that, through analysis, proved essential for sustained cooling. Each design element was evaluated against environmental, regulatory, and operational constraints. This revealed that a modular insulated structure provided the most reliable and cost-effective performance in field conditions. Thanks to a PCM pack and a peltier cooling module, our device will have an autonomy of over 35 hours.

The conclusion reached after the analysis in this report is that the proposed system is technically feasible and has strong potential to meet the needs of insulin-dependent people in low-income regions. This conclusion is supported by quantified thermal performance, a projected unit cost well below the £80 limit, and risk assessments confirming manageable regulatory and operational challenges. Overall, initial analysis indicates that the project is achievable within realistic engineering, budgetary, and regulatory boundaries while offering a practical and durable solution for maintaining insulin safety during extended travel.

2 Introduction & Project Scope

Protein-based medicines, such as insulin, are essential for managing chronic conditions like diabetes. Many must be kept between 2–6 °C to retain their efficacy [1]. However, in low- and middle-income countries, access to reliable refrigeration is a major challenge. Electricity supply is highly unreliable; nearly 1 billion people rely on healthcare facilities with inconsistent or no power. Cold chain infrastructure gaps further limit proper storage, transport, and monitoring. Meanwhile, demand is extremely high with over 60 million people with type 2 diabetes and 9 million with type 1 relying on insulin globally, yet many in LMICs still lack access.

This project aims to address these challenges by designing a portable, energy-efficient cold storage device for transporting and storing insulin and similar medicines in remote or resource-constrained environments [2]. The goal is to create a device that maintains a suitable internal temperature for 72 hours without electricity, in ambient conditions ranging from 10–30 °C. To remain competitive and accessible to LMICs, production costs must stay below £80, with a minimum lifespan of six years without major component replacements.

Libya is a key target market due to rising diabetes rates. The International Diabetes Federation reports that 15.8% of Libyan adults have diabetes, with many cases undiagnosed due to underfunded healthcare [3]. This burden is only expected to grow. Combined with political instability and a poverty rate nearing 40% [4], the need for safe and reliable insulin transportation is significant. Frequent power outages and high ambient temperatures

hinder proper storage, while limited rural healthcare access forces long travel distances, reinforcing the need for portable insulin solutions [5].

Our portable device addresses these issues by maintaining insulin stability during travel and periods of unreliable electricity. With its combined PCM and peltier cooling system, it remains effective across environments where conventional refrigeration is unreliable or unavailable. The expected project budget is around £70,000, covering up to six mid- to senior-level engineers at hourly rates between £40 and over £100, for a combined 600 hours of work. A surplus margin accounts for unforeseen circumstances. During the first year of production, 3,000 units will be manufactured at around £60 each, assuming £70,000 fixed costs, this yields an estimated £100,000 total profit.

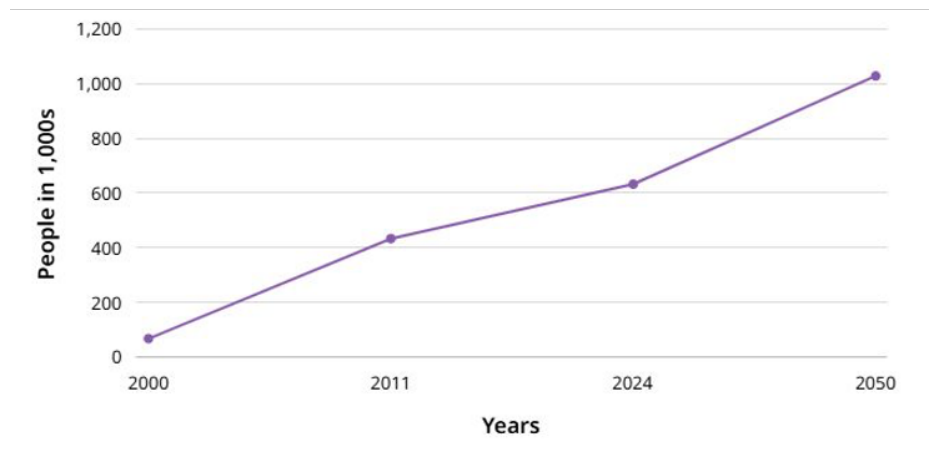


Figure 1: Diabetes Cases in Libya

The production and implementation of an Insulin carrying device introduces a number of risks that need to be addressed and considered. These risks span further than just technical risks associated with the device but include regulatory risks, legal risks, ethical risks, operational risks, environmental risks, economic risks, and social risks. Typically, medical devices have a long approval time, with a device like ours likely being 1-2 years, as it falls under an EU class 2 [6]. However, this time can be reduced by ensuring that production complies with ISO standards such as ISO14791 hence mitigating regulatory risks.

A wide variety of standards will have to be adhered to in order to ensure it is safe, transportable, and legally permitted to be sold and distributed within Libya. These include being held liable for patient safety if the cooler fails to maintain a temperature window of 2-6 °C. This also comes under ethical considerations about non-maleficence, as exposing insulin to inappropriate temperatures can reduce its potency and place patients at risk [7]. Furthermore, the battery must be UN38.3 compliant to be deemed safe for transport according to UNICEF [8]. In addition, the product must be contractual and warranty compliant as any claims made, such as a promised lifespan, must be adhered to. Otherwise, failure to do so is legally actionable.

A further aspect of legal standards that need to be adhered to is intellectual property rights (IP). The product must avoid infringing on existing designs that are protected by patents or registered designs. To mitigate this, components and design features should be open-source and come under freely licensed technology. Furthermore, securing design rights, patents, and copyright early on for any novel aspects of the device avoid infringement

from competitors [9]. In addition, NDAs can be obtained to prevent the design from being stolen. Ethical risks also relate to the responsibilities associated with the device and the duty of care that it must uphold for medical use. It must remain affordable to ensure equitable access for those living in poverty so that users can benefit from the technology, regardless of their finances or accessibility to healthcare provisions. Furthermore, by acknowledging the lack of environmental infrastructure within LMICs, like Libya, the use of materials and components cannot contribute to waste burdens [10].

Technical risks outline the device's ability to maintain performance as a tight budget can lead to lower quality components and materials. This may lead to poor long term reliability, quick battery degradation, or poor cooling. These can be mitigated by careful resource selection, durability testing, implementing regular diagnostics of the device and testing the thermal performance in realistic environmental conditions [8][11]. Operational risks include user handling errors during transportation and distribution within LMICS due to factors such as language barriers and inexperience when it comes to handling medical grade devices. Therefore, by using a simplistic design these risks can be lowered. Within a hot climate where blackouts are frequent and electricity generation is unstable, environmental risks prove to be a big factor as the device may incur charging interruptions and degradation [8]. A possible mitigation could be the use of an off-grid diesel generator to recharge the battery of the product. Further environmental risks include severe weather conditions such as sandstorms or torrential rainfall. By testing the product to ensure it attains an IP rating of IP65, it can provide complete protection from dust or dirt entering the enclosure and is able to protect the product against water jets from any directions[12]. This rating is achievable with proper sealing techniques and adherence to established enclosure design standards.

Economic risks arise if the device proves to be too expensive for distribution within LMICs. By optimising the design for simplicity and partnering with NGOs such as *UNICEF* or *Médecins Sans Frontières*, which have specialists in crisis regions, these risks can be mitigated. Furthermore, social risks may include communities within LMICs, such as Libya, being hesitant to use new medical technology due to cultural norms and beliefs and health workers preferring proven methods or devices already in use [13]. This issue can be alleviated by running campaigns endorsed by local charities and providing healthcare workers with training prior to deploying the device to emphasise its ease of use. Therefore, all selected parameters of our design ensure that risks are minimised without hindering functionality.

3 User Requirement Specifications

The User Requirement Specifications (URS) were established based on the essential medical needs related to the safe transport and storage of insulin within the required temperature range [7]. Potential conditions of use for the product were taken into account, including environmental constraints, as well as the target users, their limited resources and the need for simple handling of the device.

In addition, international medical, environmental, and electrical standards were consulted to ensure the product's regulatory compliance. Reliability and lifespan requirements were defined in line with the context of low-and middle-income countries (LMICs). The available budget and manufacturing feasibility also influenced the choice of materials and the complexity of the system. The initial design explored in this report complies with the defined requirements and the budget constraint of £80.

<i>User Requirement Specifications</i>		
<i>Category</i>	<i>Specification</i>	<i>Priority (1–3)</i>
1. Function	1.1.a. The compact device must store the insulin between 2–6°C for 72h.	1 – High
	1.1.b. The insulin must never be exposed to direct sunlight.	1 – High
	1.2.a. The compact design must be able to carry at least 50 doses of insulin.	2 – Medium
	1.3.a. The compact fridge must be able to carry insulin in the form of vials and epi pens.	1 – High
2. Environmental	2.1.a. Manufacturer and production must be ISO 14001 Environmental Management compliant.	1 – High
	2.2.a. Materials must be UV-resistant, corrosion resistant, and durable to maintain a lifespan of 6 years.	1 – High
	2.3.a. The product must have an IP rating of IP 65.	1 – High
3. Ergonomics	3.1.a. The fridge's total mass must not exceed 8 kg.	1 – High
	3.2.a. The fridge must include an adjustable padded strap and a sturdy handle.	1 – High
	3.3.a. The fridge shall utilise a latch, lever, or magnetic closure requiring around 20 N hand force.	2 – Medium
	3.4.a. Locking and unlocking process must be achievable via a single continuous movement.	3 – Low
	3.5.a. The device exterior must feature a digital or colour-coded LED/LCD indicator visible at 2 meters in full daylight.	1 – High
4. Product Quality	4.1.a. The product must be able to last 72 months.	1 – High
	4.1.b. The product must endure its full lifespan, including two years in harsh weather conditions.	2 – Medium
	4.2.a. The product must be able to withstand falls up to 1.8 meters.	1 – High
	4.3.a. The product must retain basic functionality during non-critical component failure, and users must be able to replace such components with minimal resources.	1 – High
	4.3.b. The device must maintain safe operation under critical component failure through an appropriate backup provision.	1 – High
5. Safety	5.1.a. Battery pack shall comply with UN 38.3 transport tests; chargers should include fuses.	1 – High
	5.2.a. All internal surfaces shall be sanitise-friendly and antimicrobial, consistent with ISO 13485 standards.	1 – High
	5.3.a. Clear manufacturing and expiry information must be visible on the display.	2 – Medium
6. Aesthetics	6.1.a. All external casing and handle surfaces must be made of non-porous, easily cleanable material.	2 – Medium
	6.2.a. All symbols shall follow ISO 7001 or IEC 60417 graphical standards, and text if used must appear in English and one optional local language.	3 – Low
7. Cost	7.1.a. The cost of the design must remain under £80.	1 – High

4 Concept Development

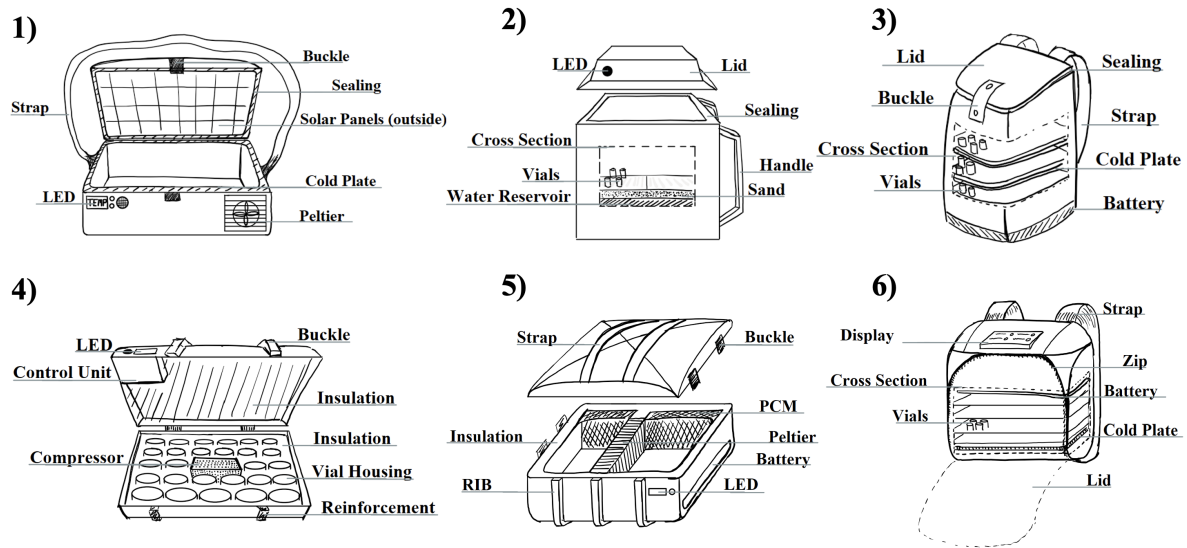


Figure 2: Developed Concepts

4.1 Concept 1 - Case with solar panels and peltier cooling

The first design proposes the use of a peltier module in combination with a solar panel. Although this design removes the need to draw electricity from an external power supply, the solar panel is not able to provide the continuous DC power supply required by the peltier module. This would require a substantially larger battery to maintain the 2-6°C range for 72 hours (URS-1.1.a). This added mass would likely push the weight beyond the desired 8kg (URS-3.1.a) and reduce portability (URS-3.2.a). Affordable solar panels also do not meet the 72 month lifespan (URS-4.1.a, 2.2.a) and upgrading to premium ones would significantly exceed the cost cap of £80 (URS-7.1.a). Whilst this design is able to deliver an energy independent solution, it is not able to do so within the budget or ergonomic constraints.

4.2 Concept 2 - Zeer Pot

Concept 2 was inspired by the Zeer Pot, an ancient evaporative cooling method. It consists of two clay pots, an inner and outer pot, with the gap between being filled with moist sand and water kept in a reservoir below. Water evaporates through the porous outer pot, cooling the inner chamber, providing a passive cooling method that does not require electrical components. This makes the concept an energy efficient and cost-effective option supporting (URS-7.1.a). However, cooling is entirely dependent on ambient humidity and airflow, meaning it cannot consistently maintain temperatures. Therefore, maintaining a temperature of 2-6°C for 72 hours (URS-1.1.a) is not guaranteed and is less precise than active electronic cooling methods. These temperature fluctuations could compromise the insulin (URS-1.1.b). Furthermore, the design has notable drawbacks as it is heavy and fragile, meaning that it would not withstand a drop of 1.8 m (URS-4.2.a). Therefore, despite its simplicity and price, it is not practical for insulin transport.

4.3 Concept 3 - Backpack with peltier module and PCM

Concept 3 is an insulin-cooling backpack that uses a hybrid system combining a peltier module with phase change material (PCM) panels designed to melt at approximately 5°C. This supports stable storage within the 2-6°C range and protects against short-term power loss (URS-1.1.a, 1.1.b). The insulated compartment contains organised vial trays and LED lights to indicate temperatures (URS-3.5.a, 5.3.a). The system is powered by a 12v battery and backpack straps enhance comfort and portability (URS-3.2.a). The main concern with this design is the limited physical protection provided by a soft backpack which may prevent it from withstanding a 1.8m fall (URS-4.2.a).

4.4 Concept 4 - Case with compressor cooling

Concept 4 is a rigid hard-shell box with a hinged lid secured by 2 latches. A compressor cools the vials maintaining 2-6°C and polyurethane boards act as insulation (URS-1.1.a). The compressor is able to provide efficient and reliable refrigeration controlled by sensors. A battery ensures it can operate with no external power supply allowing it to maintain the desired temperature for the 72 hour time frame. However, the compressor is a heavy component and when combined with the battery may lead to a weight in excess of 8kg (URS-3.1.a). The mechanical complexity and vibrations from the compressor may also compromise durability (URS-2.2.a) and ease of maintenance (URS-5.2.a).

4.5 Concept 5 - Hard case with hybrid peltier and PCM module

Concept 5 is a hard case with ribbing to protect against impacts (URS-4.2.a) and uses a hybrid cooling system combining a peltier module with PCM bricks. When connected to an external power supply it relies solely on the peltier module, in turn 'recharging' the PCM bricks. During independent operation, the PCM bricks maintain the 2-6°C range, with a small battery able to activate the peltier if temperatures rise above the safe window (URS-1.1.a, 4.1.a). Aerogel insulation further improves thermal efficiency. However, the combination of the battery, peltier module, PCM bricks and ribbing may result in excessive weight, potentially exceeding the 8 kg limit (URS-3.1.a) whilst adding undesirable complexity that could impact maintenance and durability (URS-2.2.a).

4.6 Concept 6 - Hard case cooler within backpack

Concept 6 presents a backpack containing a rigid-cooled container to protect insulin vials from impacts (URS-4.2.a). LED indicators display the internal temperature, insulin status and battery life. LED lights offer lower power consumption compared to OLED or LCD alternatives reducing energy consumption and increasing independent cooling time (URS-4.1.a, 5.3.a). A peltier module cools and maintains the vials at the required temperature (URS-1.1.a). However, the heavy battery required to power a cooling system solely dependent on a peltier module increases the overall weight and may compromise ergonomics, despite the backpack's practical form factor (URS-3.1.a, 3.2.a).

5 Selected Solution

Design Decision Matrix : Insulin Fridge																	
Criteria	Weight (1-3)	Benchmark Solution		Concept 1 : Case with Solar Panels + Peltier Coolinhg		Concept 2 : Zeer Pot		Concept 3 : Backpack with Peltier module and PCM		Concept 4 : Case with Compressor Cooling		Concept 5 : Hard Case with hybrid Peltier +PCM		Concept 6 : Hard Case cooler within backpack		NEW SOLUTION	
		R - Rating (1-5)	WS - Weighted Score	R	WS	R	WS	R	WS	R	WS	R	WS	R	WS	R	WS
Function	2.75	3.75	10.5	3.25	9.25	2.50	7	4.50	12.50	3.75	10.75	5	13.75	4.75	13	5	13.75
Environmental	3.0	1.30	3.90	2.67	8	1.67	5	3.33	10	3	9	3	9	3.33	10	4	12
Ergonomics	2.40	3.80	8.80	3.80	8.60	2.80	5.60	4.20	11	3.20	7.60	3.80	9.80	4	9.20	5	12
Product Quality	3.0	1.20	3.60	2.80	8.40	1.60	4.80	3	9	2.40	7.20	4	12	2.80	8.40	4.20	12.6
Safety	2.67	2.30	6.33	4	10.33	3	8.33	3.33	9	2.67	7	3.33	9	2.67	7	5	13.33
Aesthetics	1.50	2.50	4.50	4	6.50	2.50	3.50	2	2.50	4	6.50	3.50	5.50	3.50	5.50	2.50	4.50
COST	3	3	9	2	6	5	15	2	6	3	6	2	6	3	9	3	9
DFM	2	3	6	3	6	4	8	3	6	3	6	3	6	3	6	3	6
TOTALS		20.92	52.73	25.52	63.08	23.07	57.23	25.37	66.00	25.02	63.05	27.63	71.05	27.05	68.10	31.07	83.18
Percentage Totals			52.00		62.09		56.33		64.96		62.06		69.93		67.03		81.87

(Green $\geq 70\%$, 40% $\leq$ Yellow $< 70\%$, Red $< 40\%$ of maximum weighted score (15))

Figure 3: Design Decision Matrix for Concept Selection

A design decision matrix was compiled to determine which concept best met the User Requirement Specifications. To improve readability within the report, individual sub-criteria were grouped into their main URS categories and mean weighted scores were used for each section. Instead of pursuing a single concept, the highest rated components were merged into one cohesive design. All concepts were evaluated against the criteria in the URS. The features that stood out most were the peltier cooling system combined with PCM, which best satisfied the cooling performance required for maintaining insulin between 2-6 °C for 72 hours (URS 1.1.a), while also ensuring the insulin is never exposed to direct sunlight (URS 1.1.b). EPS insulation and a POM plastic casing received the highest ratings for durability, moisture resistance, and long term environmental performance (URS 2.2.a, 2.3.a), while also contributing to keeping the total mass below the required limit of 8 kg (URS 3.1.a). The resulting design therefore incorporates a peltier-PCM cooling system, EPS insulation, and a POM outer casing. Additional features include an LED indicator display and an emergency buzzer, which ensure visible temperature and safety status feedback (URS 3.5.a, 5.3.a). The full system remains within the cost target of under £80 (URS 7.1.a). The final New Solution corresponds to the CAD model and integrates the updated components identified through this evaluation. Ultimately it scored higher in 6 out of 8 categories, and achieved the highest overall weighted score.

Figure 4 shows the CAD drawings of the current design stage. The fridge's exterior consists of a single moulded or milled POM casing, a material chosen for its UV resistance, corrosion resistance, and durability in harsh environments (URS 2.2.a, 4.1.b). Surrounding almost the entire internal volume is a 50 mm layer of EPS insulation, selected for its low weight, high thermal resistance, moisture control capability, and cost effectiveness (URS 6.1.a, 2.2.a). Calculations (see appendix) confirm that 50 mm insulation corresponds to a reasonable battery size. A 20 Ah battery housed beneath a POM bottom plate powers the peltier module and display, meeting battery safety requirements (URS 5.1.a). PCM packs are mounted inside the lid, and temperature sensors regulate cooling performance (URS 4.3.a).

Several PCMs were considered, finally selecting Phase 5TM. It provides thermal protection for products stored

or shipped within the 2–8°C range. In liquid form, they prevent temperatures from dropping below 2°C, and in solid form, they protect against exceeding 8°C [14], which is in accordance to URS 1.1.a.

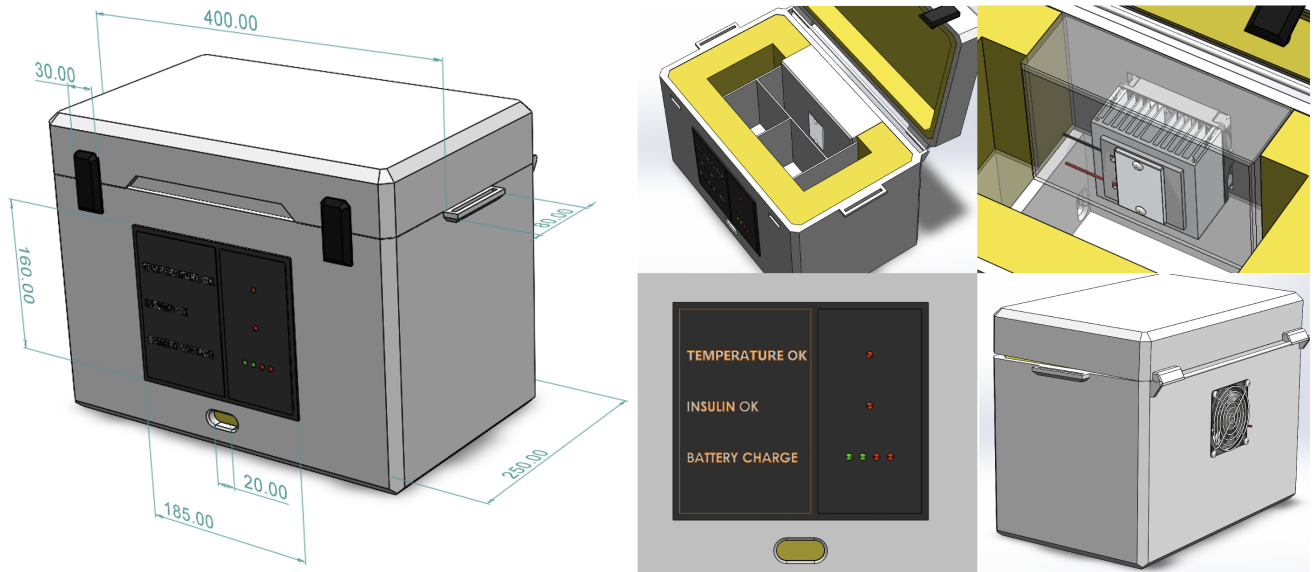


Figure 4: CAD Renders of Selected Solution

6 Feasibility Assessments

From a technical perspective, the further developed solution requires the URS to be altered. The battery size required for the requirement specification point 1.1.a exceeds slightly that of a reasonable weight. Assuming an ambient temperature of 30°C and an internal temperature requirement of 4°C cooled with a peltier and PCM bricks, with 5 cm of EPS insulation, the minimum battery capacity to function for 72 hours non-stop is of 48 Ah. Hence, the URS must be modified to only require 40 hours of autonomy. With this modification, a 20 Ah battery which weighs roughly 2-3 kg is chosen. With the peltier system using 88% of battery storage capacity, this provides a 12% battery storage margin for electrical control systems. It also accounts for potential battery degradation overtime leading to reduced capacity, ensuring an operational time for at least 5 years.

The design, as per criteria 4.3.a. in the URS, the design must be easy to maintain with critical components being easy to replace by person with limited engineering experience. With respect to making components easy to replace, the current solution is said to be ‘modular’. This means that all parts can be removed with relative ease and then be replaced. Examples include the peltier unit itself, PCM panels, battery, display and insulation. Some of these will be friction fitted (such as the insulation) with others being screwed in (such as the peltier unit). Therefore, with this modular setup if any of the parts fail the solution is a matter of removing the broken part and replacing it with a functional new part. The old part can then be repaired by experts and reused or simply recycled for its materials. If the product is used as intended, damage and the requirement for repairs/replacements should be minimal with it only being required in case of significant unexpected external damage.

With respect to maintenance, the product’s casing will be most affected as the internal parts are protected from the surrounding environment. Therefore, maintenance of internals should be minor with the most significant aspect being the good care of the peltier unit which can be ensured with a stable DC power supply and effective

heat management. The maintenance and repair of the casing and hinges will be the significant challenge with impacts possibly causing cracks. For minor damage this can be mitigated by taping over them as it is not critical to insulation nor is it a sterile surface that touches the vials. However, major damage will lead to the product being rendered dysfunctional, yet this will only be the case if there has been a major impact (out of the user requirements durability scope) meaning most the internals will likely also be damaged and the product is not expected to still be functional.

On a user level, the solution is designed for simple and intuitive operation. A green and red LED indicator clearly shows whether the internal temperature is within the safe range and whether the battery is sufficiently charged. This allows the device to be operated by healthcare workers, pharmacists, or outreach staff and volunteers with minimal training. The system requires no calibration during operation, reducing the risk of user error. In addition, the lightweight design ensures that the device can be transported over longer distances without excessive physical strain.

With respect to system integration, no changes to existing cold-chain workflows in Libya are required. Insulin is already stored in refrigerators at clinics or pharmacies and then transported by staff to where it is required, in accordance with standard cold-chain guidance [7][13]. The proposed device simply replaces the risky use of conventional coolers or ice packs currently used during transport. The battery-powered operation allows the device to remain functional during power cuts and throughout transport, which is particularly important in rural areas with intermittent electricity supply. As a result, the solution can be adopted directly into current healthcare distribution systems without the need for additional infrastructure or procedural changes. Finally, the project budget is respected reasonably well. Table 2 outlines a predicted bill of materials for the fridge. The figures are given for buying individual components. A total cost of £59.8 was achieved, allowing for a respectable margin when looking to sell. The manufacturing costs for this can be estimated at an added £5 per unit, yielding a total unit cost of £64.8 per unit. This leaves around £15 per unit of profit. Buying in bulk would let us take advantage of economies of scale, hence increasing potential profit.

Item	Part Name	Qty	Specification / Description	Notes	Cost	Total
1	Hardcase Outer Shell/POM Casting	1	Polycarbonate/ABS, impact-resistant, POM CAST	Injection molded	£5.00	£5.00
2	PCM pack Phase 5™	1	FGEL00051, Cryopak	Mounted in walls/lid in packs	£5.4/kg	£10.80
3	Peltier Module	1	TEC1-12706, 12V, 60W	Heatsink + fan	£20.00	£20.00
4	Lithium Battery Pack	1	20Ah LiFePO4-integrated BMS	With BMS	£10.00	£10.00
5	Temperature Sensors	2	DS18B20 / NTC 10k	Chamber + PCM	£1.50	£3.00
6	Cooling Fan	1	40–50mm DC fan	Hot side	£3.00	£3.00
7	Alarm Buzzer	1	High/low temp alerts	Lid mounted	£0.50	£0.50
8	Sterile Inner Liner	1	Medical grade plastic	Easy to clean	£2.00	£2.00
9	Charging Port	1	12V DC / USB-C PD	Power input	£1.00	£1.00
10	Insulation Foam	1	EPS	Around chamber	£3.00	£3.00
11	LEDs	6	Status indicators	Lid mounted	£0.25	£1.50
12	Wiring+Connections	1	—	—	£2.00	£2.00
			Overall Total Cost:	£59.80		

Table 1: Bill of Materials

The primary operational expense is electricity for recharging. With a 20 Ah battery (240 Wh) recharged every 40 hours of use, annual energy consumption is approximately 88 kWh. Assume an average electricity price £0.34/kWh for a small scale off-grid diesel generator, this amounts to under £30 per year. Additional running costs include periodic replacement of the peltier cooling module (every 5 years, £20) and the LiFePO₄ battery (every 5–7 years, £30), yielding an average annual maintenance cost of around £6. To fully recover total production costs, approximately 875 units would need to be sold. With a sales target of 1000 per year, this would take around 10 months. Once the fixed costs are recovered, the company can generate additional profits. This offers a promising return on investment, with clear profitability after cost recovery.

7 Sustainability Considerations

Sustainability was a key factor in assessing the feasibility of the selected solution. As the product is likely to be used in regions where waste management infrastructures are very limited, it was important to incorporate features that reduce environmental harm as much as possible. Most prominently, the modular architecture of the fridge allows for easy replacements of parts, resulting in a fridge that does not require complete disposal. Notably, the peltier, EPS insulation, and battery can all be changed with minimal effort or training. Replacing such parts allows for a circular economy approach, reducing the environmental impact of the fridge.

Furthermore, materials were also heavily assessed for their environmental impacts. Polyoxymethylene (POM) was selected due to its toughness, longevity, and resistance to change under UV light exposure [15]. Although POM is a petroleum derived plastic, its long lifespan means a lower likelihood to be needed to replace and thus offsetting carbon emissions released during production. Furthermore, expanded polystyrene insulation (EPS) has a large carbon footprint during production, however, its extremely good thermally insulating properties allows for a smaller battery to be used. Hence resulting in less energy consumption in a grid where power can be scarce. Yet in future iterations, a bio-based foam insulation should be considered, however this would not be feasible at the moment within the £80 budget.

The electrical systems were also evaluated for their environmental impact. The choice of LiFePO₄ batteries balances environmental and safety considerations. A LiFePO₄ battery has a significantly longer lifespan than other lithium ion batteries [16]. Additionally, they are much safer thermally and produce less heat, which is essential to reduce the power consumption of the refrigeration unit. As well, the energy consumption is further reduced by using a phase changing material (PCM) that passively cools the space, resulting in the peltier module only being used intermittently. This reduces the electricity demand in an already unreliable energy grid.

Finally, considering the end of the life of the product, many of the used materials are recyclable. This including the POM casing, EPS insulation, Aluminium heat sink, and the LiFePO₄ batteries. The modular layout of the fridge will simplify the disassembly, and the use of few permanent fasteners and composite materials reduces the amount of energy required to separate the materials to recycle. However, recycling infrastructure in Libya is very limited, the modular design will allow local NGOs to centralise end of life collection and transport components to the appropriate recycling centres. Overall, as the device uses several manufacturing materials, its long service life, low energy demand, and repair-ability demonstrate that the environmental impact has been considered throughout the design. These decisions collectively support URS category 2, which requires ISO 14001-compliant manufacturing, durable UV-resistant materials, and an environmentally robust design [17].

8 Recommendations & Conclusions

Assessing the current prototype against the established user requirements indicates that continued development of the compact fridge is both viable and worthwhile. The proposed solution achieves a weighted score of 92% against the URS, significantly higher than all other concepts. However, alterations will still need to be made. Durability remains one of the most challenging and critical aspects of the design, particularly the requirement for the device to retain functionality for 72 months and survive a 1.8 m drop. Although physical testing cannot be conducted, the current design is unlikely to meet this criterion. Its high mass means that an impact from such a height could crack the POM housing, damage electronic components, and compromise the insulin stored inside. To address this, we recommend incorporating rubber ribbing or protective padding around the device's edges and corners to better absorb impacts.

Despite these concerns, the design demonstrates several advantages, most notably the projected battery life exceeding 35 hours, ensuring reliable operation between charges. Although this is not as high as anticipated, this time-frame still provides enough time between charging stations. However, the lithium ion battery shall be subjected to further testing under United Nations Regulations [18]. Additionally, the combination of a PCM and peltier cooling module provides redundancy; if one system fails, the other can operate as a backup to maintain safe storage conditions. We also recommend use of a storage rack within the insulin storage compartments to prevent spillage and rattling of vials. According to the CAD renders, there are 5 compartments for insulin storage in the design, of which 3 unique designs for the storage rack would be needed. With the implementation of the storage racks, the maximum number of insulin vials that could be transported would be 63.

Looking ahead, further work is required to reinforce structural integrity and enhance long-term durability. As well, more thorough thermal calculations will be carried out to establish the fundamentals of the software behind the display. The electronics of the fridge will also need further attention. Additional research into commercialisation pathways, relevant standards and codes of practice, and detailed manufacturing processes will also be essential as development progresses. Finally, although a budget of £80 per unit is feasible, it hinders the quality of materials that we source. Hence, it would be beneficial to increase the budget to £100 per unit, allowing for a more generous profit margin and higher quality resources.

We recommend applying a systems approach, ensuring an interconnected set of components that must work together collectively. This includes cooling mechanism, power supply, insulation, temperature sensors, and housing that must all be compatible. Breaking the goal into subsystems allows for a more structured approach for the design, ensuring a safe, efficient, and dependable medical device.

9 Project Schedule

Figure 5 displays a Gantt chart displaying the general project schedule. Given the absence of physical prototyping, a technical field study will be conducted during the first two weeks of January, in parallel with the initial CAD and thermal and electrical analyses and modelling. This should avoid any impact on the project's critical path. It will enable the choices of materials, durability, and power supply systems to be validated using real data. This approach ensures our final concept can be delivered by end of February without major late modifications.

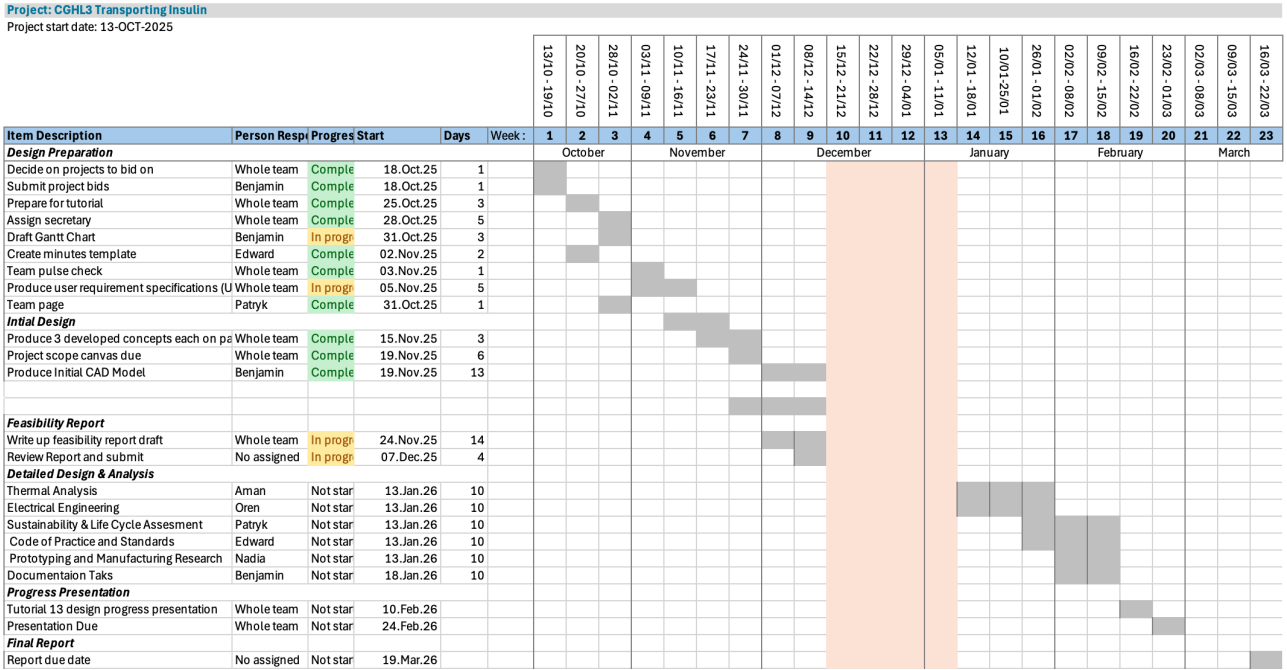


Figure 5: Project Gantt Chart

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11 Appendices

11.1 Appendix 1: Battery Capacity Calculation

For an insulin carrier case with dimensions of $0.2m \times 0.2m \times 0.2m$.

The surface area A is given by:

$$A = 6 \times 0.2^2 = 0.24 \text{ m}^2$$

Internal temperature: $T = 4^{\circ}\text{C}$, Ambient temperature: $T_0 = 30^{\circ}\text{C}$, Difference in temperature: $\Delta T = 26^{\circ}\text{C}$

EPS Thermal conductivity: $k = 0.035 \text{ W/m}$ [19]

Assuming an EPS thermal insulation thickness d of 5 cm, Total conductive heat through insulation is [20]:

$$\dot{Q} = \frac{kA\Delta T}{d} = \frac{0.035 \times 0.24 \times 26}{0.05} = 4.37 \text{ W}$$

The design must operate for at least 40 hours without external power. For a 40 hour blackout period:

$$t = 40 \times 60^2 = 144,000\text{s}$$

Total conductive heat energy through insulation:

$$Q_{\text{total}} = \dot{Q} \times t = 4.37 \times 144000 = 629 \text{ kJ}$$

Take the PCM material with a specific latent heat of 200 kJ/kg during phase change and a mass of 2kg [21]:

$$\dot{Q}_{\text{PCM}} = \frac{m_{\text{PCM}} \times L_{\text{PCM}}}{t} = \frac{2 \times 200 \times 10^3}{144000} = 2.78 \text{ W}$$

The total heat energy removed is equal to the sum of cooling contributions from both the PCM and the peltier cooling system.

$$\dot{Q}_{\text{Peltier}} = \dot{Q}_{\text{total}} - \dot{Q}_{\text{PCM}} = 4.37 - 2.78 = 1.59 \text{ W}$$

A peltier cooling system typically has a Coefficient of Performance (COP) of 0.2 - 0.4 depending on real-world factors. Assume a COP of 0.3 for the whole system.

$$\dot{E}_{\text{Peltier}} = \frac{\dot{Q}_{\text{Peltier}}}{\text{COP}} = \frac{1.59}{0.3} = 5.3 \text{ W}$$

Battery capacity (Wh) required:

$$Wh = \dot{E}_{\text{Peltier}} \times t = 5.3 \times 40 = 212 \text{ Wh}$$

We will be using a battery of $V = 12 \text{ V}$. The battery capacity in Ah:

$$Ah = \frac{Wh}{V} = \frac{212}{12} = 17.6 \text{ Ah}$$

Hence, a 12 V 20 Ah LiFePO₄ battery will be used, providing ample margin over the required 17.6 Ah.

11.2 Appendix 2: Storage capacity calculation

The dimensions of the storage compartment in length $\times$ width $\times$ height is $300 \times 180 \times 160 \text{ mm}$.

The average dimensions for a single insulin vial is $\varnothing 24.0 \times 45 \times 1.00 \text{ mm}$ [22].

Accounting for space between vial grids on the rack and the dimensional property of the storage compartments, assume each insulin vial takes up 50 mm^2 over the total area.

The number of layers of racks is $\frac{160}{3} = 3.56 \approx 3$.

The number of grids on each rack is $\frac{300 \times 180}{50^2} = 21.6 \approx 21$.

Hence the total number of insulin vials that can be carried is given by $3 \times 21 = 63$.